Journal of Medical Entomology, 54(4), 2017, 1031–1036
doi: 10.1093/jme/tjx047
Advance Access Publication Date: 13 March 2017
Vector Control, Pest Management, Resistance, Repellents Research article

Efficacy of Bendiocarb Used for Indoor Residual Spraying for Malaria Control in Madagascar: Results With Local *Anopheles* Species (Diptera: Culicidae) From Experimental Hut Trials

Sanjiarizaha Randriamaherijaona,[1,2,3] Thiery Nirina Jean Jose Nepomichene,[1,2] Jade Assoukpa,[1,4] Yoann Madec,[4] and Sébastien Boyer[1]

[1]Unité d'Entomologie Médicale, Institut Pasteur de Madagascar, BP 1274 Antananarivo, Madagascar (sanji@pasteur.mg; Jthiery@pasteur.mg; assoukpa.jade@yahoo.fr; seboyer@pasteur.mg), [2]Ecole Doctorale Sciences de la Vie et de l'Environnement, Université d'Antananarivo, Madagascar, [3]Corresponding author, e-mail: sanji@pasteur.mg, and [4]Unité d'Epidémiologie des Maladies Emergentes, Institut Pasteur 25-28, rue du Docteur Roux, 75015 Paris, France (yoann.madec@pasteur.fr)

Subject Editor: William Reisen

Received 20 July 2016; Editorial decision 21 January 2017

Abstract

To control malaria in Madagascar, two primary vector control interventions are being scaled up: insecticide-treated nets and indoor residual spraying of bendiocarb, which was implemented in the Malagasy Central Highlands in 2009. The current efficacy of bendiocarb against *Anopheles* species was evaluated in a small-scale field trial. An experimental hut trial comparing the effectiveness of bendiocarb sprayed on five substrates (cement, wood, tin, mud, and vegetative materials) was carried out against *Anopheles* species in two study sites located in the eastern foothills of Madagascar. No significant difference was detected in either exophily or blood-feeding rates between treated and untreated huts. The mortality rate was significantly greater in treated huts compared to untreated huts. Efficacy up to 80% was found for 5 mo posttreatment. Although effective, bendiocarb has been used for 7 yr, and therefore an alternative insecticide may be needed to avoid the emergence of resistance.

Key words: Indoor residual spraying, bendiocarb, residual activity, experimental hut, Madagascar

The World Health Organization (WHO) estimates that 215 million cases of malaria occurred globally in 2014 (uncertainty range: 149–303 million), leading to 438,000 deaths (uncertainty range 236,000–635,000). Most deaths (90%) occurred in Africa, and most deaths (69.8%) were in children under 5 yr of age (Bhatt et al. 2015, WHO 2014).

Primary control of malaria vectors is achieved through two main methods: insecticide-treated nets (ITNs) and indoor residual insecticide spraying (IRS; WHO 2014). Indoor residual insecticide spraying has proven to be effective in rapidly controlling malaria transmission, provided that most houses (>80%) in targeted communities are treated (WHO 2006, Griffin et al. 2010). Majority of insecticides used in IRS function through repelling mosquitoes from entering houses and by killing female mosquitoes that are resting inside houses after having taken a bloodmeal (Pluess et al. 2010). Both ITNs and IRS show a high degree of personal protection and a mass mortality effect at high coverage against susceptible mosquitoes (Pluess et al. 2010). Greater numbers of insecticides are available for IRS than used to treat ITNs. Indeed, pesticide chemical classes (organophosphates, organochlorines, carbamates, and pyrethroid) are available for IRS spraying (WHO 2007), allowing for better resistance management in vector control programs (Pluess et al. 2010).

In Madagascar, IRS began in 1949 with DDT (dichlorodiphenyltrichloroethane) to eliminate malaria (Randriantsimaniry 1995). Between 1955 and 1970, this vector control activity was intensified (Curtis and Mnzava 2000). Malaria cases were briefly dissipating at the beginning of the 1970s and malaria vectors, especially *Anopheles funestus s.l.* Gilles disappeared. In 1986, a re-invasion by *An. funestus s.l.* was noted and malaria cases increased in the Central Highlands area of Madagascar (Lepers et al. 1991, Rakotomanana et al. 2001). In 1988, a national program that was implemented to control the spread of malaria cases based on accessibility to chloroquine and IRS with DDT aimed at foci where most cases were found. From 1993 to 1998, DDT continued to be used to control the increasing distribution of the primary malaria vector *An. funestus s.l.* (Romi et al. 2002). In 2003, another major malaria vector, *An. arabiensis* Patton, was found to be resistant to DDT, but

© The Authors 2017. Published by Oxford University Press on behalf of Entomological Society of America.
All rights reserved. For Permissions, please email: journals.permissions@oup.com

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018



Fig. 1. Experimental huts. Saharevo station.

still susceptible to pyrethroids (Ratovonjato et al. 2003). In 2005, the Malagasy National Malaria Control Programme (NMCP) changed its approach and used pyrethroid insecticides for IRS (Ratovonjato et al. 2014). Since 2009, the Malagasy NMCP decided also to implement IRS using the carbamate bendiocarb in the Central Highlands. The use of different classes of insecticides was intended to delay the development and spread of insecticide resistance in malaria vectors (Guillet et al. 2001, Hougard et al. 2003). Bendiocarb while highly effective, has a short residual activity (duration of 2–6 mo when used for IRS) and is more expensive than pyrethroids and DDT (WHO 2012).

The objective of the current study was to evaluate the efficacy of bendiocarb to induce exophily (increase escape), reduce blood-feeding, and increase mortality against natural populations of *Anopheles* using an experimental hut approach.

Materials and Methods

Study Sites and Period of Survey

Two study sites, Saharevo (SRV; 18° 51′12.9″ S, 48° 07′48.8″ E) and Ambohitranivo (ABN; 18° 50′52.71″ S, 48° 14′24.17″ E), were selected. Both sites are located on the eastern edge of the Malagasy Central highlands, 100 km east of Antananarivo, in the Moramanga District, Madagascar. These villages are located within a rice-growing area with *Anopheles* densities heavily dependent on rice planting cycles. From October 2003 to September 2004, an entomological investigation in SRV reported the presence of four malaria vectors: *An. funestus s.l.*, *An. mascarensis* de Meillon, *An. gambiae ss.* Giles, and *An. arabiensis*. *Anopheles funestus s.l.* was responsible for ∼75% of the plasmodium transmission that occurred from February to May during the period of study (Andrianaivolambo et al. 2010). WHO susceptibility tests with *An. mascarensis* in 2012 showed no resistance to bendiocarb 0.1% in both study areas (SR, unpublished data). The current entomological survey was done in experimental huts from June 2013 to February 2014.

Experimental Huts

The study was conducted in experimental huts (Fig. 1) modified from a design based on WHO specifications for standard evaluation of insecticide-treated nets and IRS (WHO 2006). In each study site, 10 experimental huts were built with different wall types representing the main building materials used in Madagascar and with floor made of cement.

The wall substrates were the following:

- Mud: local red clay used for house building in rural areas of the Malagasy Central Highlands at a thickness of about 10 cm.
- Cement: mixture of cement and sand about 5 cm thick.
- Wood: pine planks.
- Tin: galvanized iron sheets classically used in Malagasy habitat construction at 0.30 cm thick.
- Vegetative: stems of *Ravenala madagascarensis* (Sonn.)

For mud substrate, the soil was different for each site and 4 d were needed for drying; walls were treated one year after experimental hut construction.

Experimental huts located in SRV were built in the valley, about 3 m from rice fields and far from the village. Experimental huts located in ABN were built on a hillside (20 m from the rice fields in the valley and 50 m from the village).

Experimental huts were 3 m long and 2 m wide and surrounded by a moat to prevent the entry of ants and spiders. A plastic tarpaulin ceiling was stretched beneath the roof to protect sleeping residents from the heat and to facilitate the capture of mosquitoes. Mosquitoes were able to enter the huts through four window slits made of iron sheet, one centimeter wide and located on the four sides of each hut. Four window exit traps were located on three sides of the hut. Each hut construction type was duplicated: one was a control hut without insecticide and the other a hut that contained walls treated with bendiocarb.

Insecticide Treatment

Bendiocarb WP 80% Ficam (Bayer (Pty) Ltd, Isando, South Africa) provided by the Malagasy NMCP was applied by qualified NMCP agents according to WHO recommendations (400 mg a.i./m^2; WHO 2013) using a Hudson sprayer (H.D. Hudson Manufacturing Company, Chicago, IL). Sprayers were equipped with new nozzles (8002E flat fan) as well as control flow valve (CFV) and calibrated just prior to applications. The walls were sprayed to attain dosages as per manufacturer's recommendation. A "red" CFV with 1.5 bar CFVs (model 98662V Viton-tipped) was used with a Hudson sprayer tank (8 liters) for the IRS.

Huts were treated concurrently in July 2013, the peak of the dry season, two months following a baseline, insecticide-free entomological survey. Baseline survey is important to ensure that 1) huts were comparable in attractiveness to the target species, 2) the target species were present in adequate numbers in the study area, and 3) experimental huts had not previously contaminated by insecticides (WHO 2006).

For each study site, 10 local adult volunteers were recruited to sleep on mats inside the 10 huts. They provided verbal consent before enrollment.

Sleepers Rotation and Mosquito Collection

At 18:30 h the sleeper supervisor opened the four window slits and the four window traps in each hut. To mimic activity of a local household, a candle was lit inside each hut. Sleepers entered the hut at 19:30 h and blew out the candle 15 min later. They slept from 20:00 to 05:30 h. At 05:30 h, after closing window traps and slits, sleepers collected mosquitoes from the floor, walls, and ceilings using glass tubes and a flashlight. Mosquitoes in window traps were collected by the supervisor. All collected mosquitoes were scored by location as dead or alive, blood fed or unfed, and identified to species (Gejbine 1966). Living mosquitoes were held in netted plastic cups in the insectary and supplied with 10% sugar solution for 24 h

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018

Table 1. Percentage of each *Anopheles* species collected in each study site

	Saharevo		Ambohitranivo	
	N	Percentage	*N*	Percentage
Anopheles squamosus Theobald	390	40.7	609	63.8
Anopheles coustani[a] Laveran	271	28.3	220	23.0
Anopheles mascarensis[a] de Meillon	190	19.8	37	3.9
Anopheles arabiensis[a] Patton	57	5.9	43	4.5
Anopheles brunnipes Theobald	24	2.5	42	4.4
Anopheles flavicosta Edwards	13	1.4	4	0.4
Anopheles sp. Meigen	7	0.7	–	–
Anopheles funestus[a] Giles	5	0.5	–	–
Anopheles rufipes Gough	2	0.2	–	–
All Anopheles species	959	100	955	100

[a] A malaria vector.

to monitor mortality. PCR identification of members of the *An. gambiae* complex was undertaken with legs or wings using a previously published protocol (Scott et al. 1993).

Volunteers were rotated among huts each night during the 10 nights of the study to minimize any effects of sleeper attractiveness. Each evaluation lasted 10 nights to allow each sleeper to rotate through all treatments at least once. Trials were repeated a total nine times over a 9-mo period, with each survey spaced ~20 d apart.

Data Analysis

The study was conducted over nine months, in 10 huts (two for each type of wall) in two villages, leading to a total number of 180 observations. Each observation consisted of 10 consecutive nights of mosquito collection. Because few *Anopheles* malaria vector mosquitoes were collected throughout, data were pooled over species for analyses and then presented separately for *An. coustani* Laveran, the most abundant malaria vector. The entomological impact of the bendiocarb treatment for each hut type was expressed in terms of induced exophily (the proportion of total *Anopheles* females found in exit traps), blood feeding inhibition (the reduction in proportion of *Anopheles* females found to be bloodfed), and mortality (the proportion of mosquitoes killed [immediate plus delayed]). When considering the substrate types or the posttreatment survey, denominators of proportion calculations were the total of collected *Anopheles* for each efficacy indicator. For each type of substrate, efficacy indicators were compared between controls and treated huts using a Kruskal–Wallis nonparametric test.

Results

Attractiveness of Huts Before Treatment

Baseline mosquito collections were performed in all huts before insecticide treatments to assess the comparability of attractiveness of each hut for mosquitoes. This study was carried out during two months (June and July in 2014). The number of *Anopheles* collected among the 20 huts for both sites did not significantly differ (mean $= 6.4 \pm 1.32$, $P = 0.79$, for ABN and mean $= 10 \pm 2$, $P = 0.63$, for SRV).

Species and Proportion of Collected Mosquito

Over the 9-mo trial, 955 and 959 *Anopheles* were collected respectively in ABN and SRV. The percentages of each Anopheles species are shown in Table 1. Four malaria vector species were collected in SRV: *An. funestus s.l.*, *An. arabiensis*, *An. mascarensis*, and *An. coustani*. However, among the five anopheles species collected in ABN, only three were malaria vectors: *An. arabiensis*, *An. mascarensis*, and *An. coustani*. For both experimental huts sites, *An. squamosus* Theobald was the most abundant *Anopheles* species, 40.7% ($n = 390/959$) in SRV and 63.8% ($n = 609/955$) in ABN. *Anopheles coustani* was the second most abundant species (28.3% [$n = 271$] in SRV and 23.0% [$n = 220$] in ABN).

Considering only mosquitoes collected in untreated huts, monthly abundance of malaria vectors was lowest during June and July, corresponding to the dry-cold season in the Malagasy Central Highlands (Fig. 2). From December to February, abundance increased for each malaria vector species.

Entomological Impacts of Treatments

Induced Exophily

Overall, the percentage of *Anopheles* found in the window traps was significantly lower in treated than in control huts in ABN ($P = 0.003$), but not in SRV ($P = 0.40$; Table 2). However, in ABN this significant difference was limited to huts made of vegetal materials ($P = 0.013$; Table 2). For *An. coustani*, which was the most common malaria vector species collected during this study, there was no significant difference between treated and control huts at both experimental sites (Supplemental file 1 [online only]).

Blood Feeding

Overall, no effect of bendiocarb was found regarding the percentage of blood-fed females at both experimental sites (respectively $P = 0.10$ and $P = 0.39$ in ABN and SRV). No significant difference was found when comparisons were stratified by substrate type (Table 2), or by sequential posttreatment survey (Table 3). The same result was observed for *An. coustani* (Supplemental files 1 and 2 [online only]).

Mortality

Over seven months posttreatment in both experimental sites, mortality rates were significantly higher in treated huts than in control huts ($P < 0.05$; Table 3) for each substrate type (Table 2). When considering only *An. coustani*, a significant higher mortality rate was recorded in treated huts than in control huts ($P < 0.05$) for each substrate (Supplemental file 2 [online only]). When the analysis was stratified by month (i.e. duration since treatment), the difference remained significant, except for the first-two months due to the low percentages of mosquitoes which entered both control and treated huts (Table 3).

Discussion

Anopheles coustani and *An. squamosus* were the most abundant species of *Anopheles* entering experimental huts in both study sites. *Anopheles coustani* is considered as a malaria vector based on parasite detection from this species in the western Malagasy Central Highlands (Nepomichene et al. 2015). *Anopheles arabiensis* and *An. mascarensis* also were previously found infected in the highlands, but with a minor role in malaria transmission compared with *An. funestus s.l.* (Gejbine 1966, Brutus et al. 2001, Le Goff et al. 2003, Pock Tsy et al. 2003). In the northern Malagasy Central Highlands, *An. coustani* and *An. squamosus* were found to be attracted by cattle (Tantely et al. 2013). Both species are considered to be mainly exophilic and zoo-anthropophagic species, which may explain the low number found blood-fed inside huts and window traps (Tantely

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018

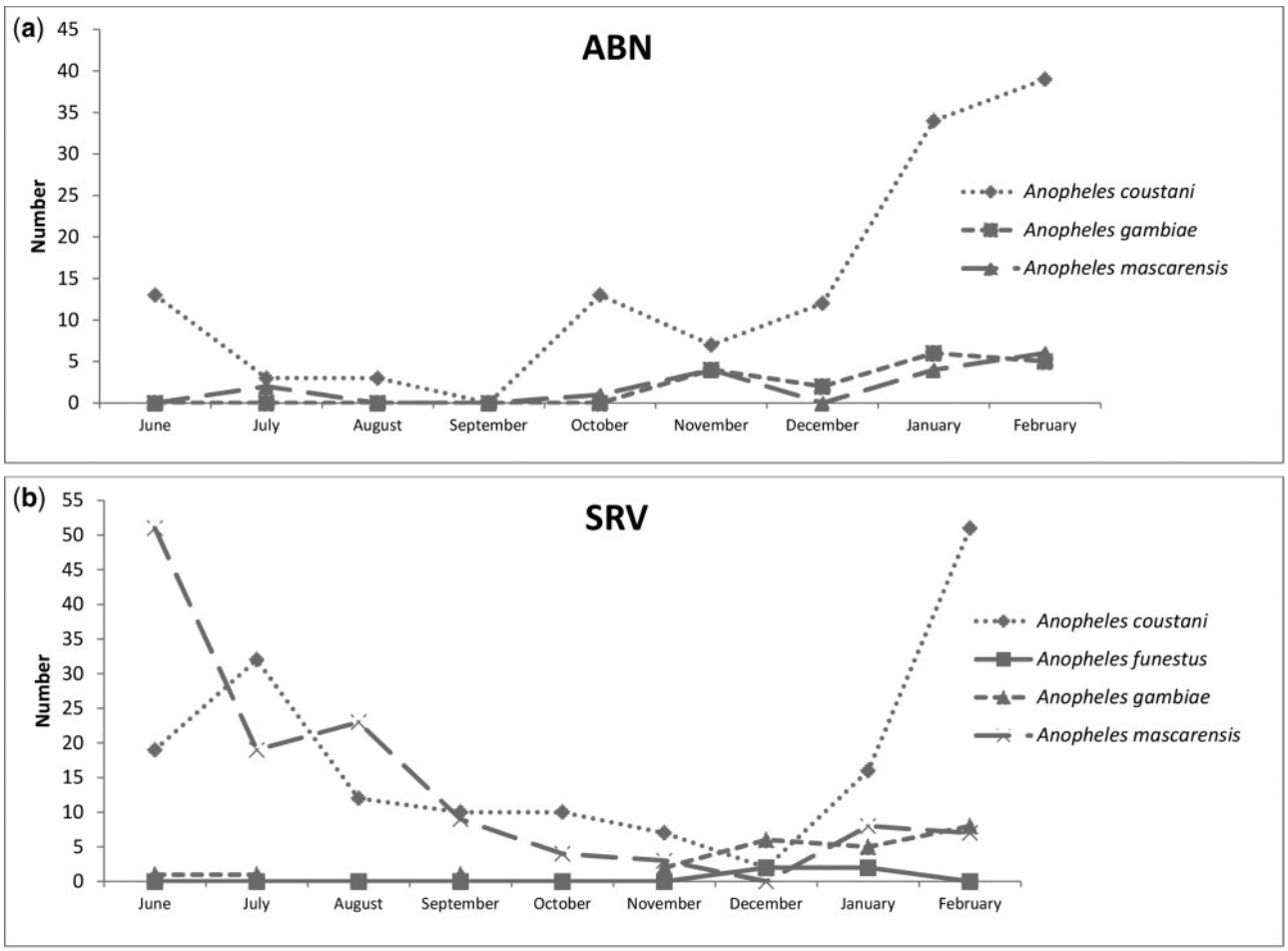


Fig. 2. Number of malaria vectors collected in untreated huts in ABN (**a**) and in SRV (**b**).

Table 2. Statistics by substrate type for all *Anopheles* species pooled by treatment status and by village (results are median [IQR])

	SRV					ABN				
	N	No. of Anopheles	Exophily (%)	Blood fed (%)	Mortality (%)	*N*	No. of Anopheles	Exophily (%)	Blood fed (%)	Mortality (%)
Control										
Wood	10	56	26 [1–55]	0 [0–10]	6 [0–12]	10	99	45 [25–50]	0 [0–4]	0 [0–20]
Cement	11	74	33 [21–83]	0 [0–3]	22 [18–38]	11	138	67 [48–100]	0 [0–1]	0 [0–45]
Vegetative	11	63	50 [29–63]	8 [3–22]	8 [0–18]	10	41	59 [50–70]	0 [0–28]	0 [0–39]
Tin	11	103	12 [0–28]	4 [0–21]	6 [0–16]	8	80	68 [57–86]	0 [0–7]	0 [0–22]
Mud	11	68	40 [10–58]	0 [0–12]	10 [3–24]	10	113	42 [16–77]	18 [3–32]	0 [0–0]
Bendiocarb										
Wood	7	93	50 [9–100]	0 [0 -12]	100 [84–100][a]	5	61	0 [0–0]	0 [0–0]	100 [92–100][a]
Cement	7	77	100 [29–100]	0 [0–7]	100 [74–100][a]	6	100	27 [4 – 85]	3 [0–5]	100 [75–100][a]
Vegetative	7	83	23 [0–100]	0 [0–9]	100 [81–100][a]	6	132	6 [0–28][a]	0 [0–7]	96 [91–100][a]
Tin	7	86	100 [4–100]	0 [0–4]	100 [88–100][a]	6	52	52 [0–92]	0 [0–0]	76 [71–95][a]
Mud	7	94	22 [0–65]	0 [0–17]	100 [68–100][a]	7	96	0 [0–28]	0 [0–1]	100 [67–100][a]

[a] Significant difference between treated and control huts ($P < 0.05$).

N are not always 11 in control huts nor 7 in treated huts because only the huts where *Anopheles* were found are considered.

et al. 2016). Residual house spraying is likely to be effective only if the target mosquito species are endophilic or at least partially endophilic, because the mosquito needs to enter houses and rest on the insecticide-treated walls for a sufficient time if to be exposed to a lethal dose (Pates and Curtis 2005). In Benin, Djènontin et al. (2010) found that the proportion of blood-fed *An. gambiae* in control huts and in huts with carbamate-treated plastic sheeting (CTPS) or IRS with bendiocarb were not significantly different. In our study, the percentage of blood-fed mosquitoes in control and treated huts also were not statistically different. Similar results were found when

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018

Table 3. Statistics by village for all Anopheles pooled by posttreatment survey (all results are median [IQR])

	SRV					ABN				
	N	No. of Anopheles	Exophily (%)	Blood feed (%)	Mortality (%)	*N*	No. of Anopheles	Exophily (%)	Blood feed (%)	Mortality (%)
Control										
1	5	51	54 [11–67]	0 [0–8]	8 [0–17]	4	51	67 [25–100]	17 [0–50]	75 [38–100]
2	4	27	100 [75–100]	0 [0–4]	79 [78–85]	2	27	50 [25–75]	0 [0–0]	0 [0–0]
3	5	22	0 [0–20]	0 [0–0]	0 [0–0]	5	22	50 [33–56]	0 [0–0]	0 [0–0]
4	5	20	29 [0–40]	20 [0–43]	0 [0–12]	5	20	50 [40–57]	12 [0–29]	0 [0–0]
5	5	28	100 [67–100]	33 [0–33]	67 [67–70]	5	28	50 [50–50]	0 [0–0]	82 [75–92]
6	5	80	27 [25–43]	8 [6–8]	16 [8–17]	5	80	63 [62–69]	6 [1–8]	6 [6–10]
7	5	136	27 [21–28]	3 [0–6]	6 [6–9]	5	136	79 [71–81]	6 [6–11]	0 [0–0]
Bendiocarb										
1	5	48	100 [100–100]	0 [0–0]	100 [75–100][a]	2	4	0 [0–0	0 [0–0]	100 [100–100]
2	5	49	100 [100–100]	0 [0–0]	100 [100–100]	3	10	0 [0–0]	0 [0–0]	100 [100–100]
3	5	58	100 [100–100]	0 [0–0]	100 [100–100][a]	5	38	0 [0 – 67]	0 [0–0]	100 [75–100][a]
4	5	20	0 [0–0]	0 [0–17]	100 [75–100][a]	5	31	0 [0–0]	0 [0–0]	100 [80–100][a]
5	5	29	100 [0–100]	0 [0–20]	100 [100–100][a]	5	59	67 [0–100]	6 [0–9]	100 [100–100][a]
6	5	79	0 [0–22]	5 [0–8]	94 [63–100][a]	5	241	12 [6–14][a]	0 [0–2]	70 [62–91][a]
7	5	150	23 [18–30]	4 [3–10]	60 [53–62][a]	5	58	50 [40–100]	0 [0–8]	67 [33–82][a]

[a] Significant difference between treated and control huts ($P < 0.05$).

comparing bendiocarb to other insecticides such as deltamethrin or alphacypermethrin applied in IRS (Akogbeto et al. 2010).

In the current study, low blood feeding rate was possibly due to the low number of mosquitoes collected inside huts and to the high mortality rate in the treated huts. However, in another study, a reduction of entry by *An. gambiae* was observed in huts treated with organophosphates but less with carbamates (Agossa et al. 2014). Additionally, Akogbeto et al (2010) reported that there was no difference in entry into experimental huts treated with fenitrothion, chlorpyriphos–deltamethrin mixture, and bendiocarb versus control huts.

In the current study, there was no significant difference between exophily in treated and control huts, further documenting bendiocarb's nonexito repellency effects. Indeed, when sprayed on cement walls and compared to other insecticides, experimental huts with bendiocarb showed a similar exophily results as controls against susceptible *An. gambiae* in Benin (Akogbeto et al. 2010). Similarly Asinas et al. (1994) reported that >75% of the mosquitoes released inside the treated huts were recaptured inside these huts, indicating that mosquitoes had not been strongly irritated and stimulated to exit by exposure to bendiocarb. Similar results were described by Guillet et al. (2001) using carbamate-treated nets in Côte d'Ivoire. In view of their nonexito repellency to adult mosquitoes, carbamates, similar to organophosphates, used for large-scale ITNs and IRS programs might have more impact than pyrethroids on mosquito populations, by inducing a "mass killing effect" thereby reducing adult mosquito densities, survival rates, and hence vectorial capacity (Curtis and Mnzava 2000).

The present results established the baseline efficacy of bendiocarb WP 80% against local *Anopheles* species on different treated substrates. These baseline data can be used for comparison to other insecticide candidates to be used in rotation in IRS in Madagascar. Candidate insecticide formulations which could be used for IRS should be tested in similar experimental huts with different substrates so that results facilitate decisions on the dosage according to each type of indoor surface at community level.

Acknowledgments

This study was supported by the Institut Pasteur de Madagascar (IPM). Professor Christophe Rogier, Director of the Institut Pasteur de Madagascar during the study, is warmly thanked. We thank mosquito collectors from Moramanga and staff from the Medical Entomology Unit of the IPM for their kind support and collaboration. We are grateful to the Malagasy NMCP for providing formulated bendiocarb and conducting spraying.

References Cited

Agossa, F. R., R. Aïkpon, R. Azondèkon, R. Govoetchan, G. G. Padonou, O. Oussou, F.D.R. Okégbo, and M. C. Akogbéto. 2014. Efficacy of various insecticides recommended for indoor residual spraying: Pirimiphos methyl, potential alternative to bendiocarb for pyrethroid resistance management in Benin, West Africa. Trans. R. Soc. Trop. Med. Hyg. 108: 84–91.

Akogbeto, M. C., G. G. Padonou, D. Gbenou, S. Irish, and A. Yadouleton. 2010. Bendiocarb, a potential alternative against pyrethroid resistant *Anopheles gambiae* in Benin, West Africa. Malar. J. 9: 204.

Andrianaivolambo, L., O. Domarle, M. Randrianarivelojosia, J. Ratovonjato, G. Le Goff, A. Talman, F. Ariey, and V. Robert. 2010. Anthropophilic mosquitoes and malaria transmission in the eastern foothills of the central highlands of Madagascar. Acta Trop. 116: 240–245.

Asinas, C. Y., C. T. Hugo, C. J. Boase, and R. G. Evans. 1994. Evaluation of selective spraying of bendiocarb (Ficam VC) for the control of *Anopheles flavirostris* in the Philippines. J. Am. Mosq. Control Assoc. 10: 496– 500.

Bhatt, S., D. J. Weiss, E. Cameron, D. Bisanzio, B. Mappin, U. Dalrymple, K. E. Battle, C. L. Moyes, A. Henry, P. A. Eckhoff, et al. 2015. The effect of malaria control on Plasmodium falciparum in Africa between 2000 and 2015. Nature 526: 207–211.

Brutus, L., G. Le Goff, L. G. Rasoloniaina, V. Rajaonarivelo, A. Raveloson, and M. Cot. 2001. Lutte contre le paludisme dans le moyen-ouest de Madagascar: Comparaison de l'efficacité de la lambda-cyhalothrine et du DDT en aspersions intra-domiciliaires. II - étude entomologique. Parasite 8: 297–308.

Curtis, C. F., and A.E.P. Mnzava. 2000. Comparison of house spraying and insecticide-treated nets for malaria control. Bull. World Health Org. 78: 1389–1400.

Djènontin, A., F. Chandre, K. R. Dabiré, J. Chabi, R. N'Guessan, T. Baldet, M. Akogbéto, and V. Corbel. 2010. Indoor use of plastic sheeting impregnated with carbamate combined with long-lasting insecticidal mosquito nets

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018

for the control of pyrethroid-resistant malaria vectors. Am. J. Trop. Med. Hyg. 83: 266–270.

Gejbine, A. 1966. Insectes Diptères Culicidae Anophelinae. ORSTOM CNRS, Paris.

Griffin, J. T., T. D. Hollingsworth, L. C. Okell, T. S. Churcher, M. White, W. Hinsley, T. Bousema, C. J. Drakeley, N. M. Ferguson, M.-G. Basáñez, et al. 2010. Reducing Plasmodium falciparum Malaria Transmission in Africa: A Model-Based Evaluation of Intervention Strategies. PLoS Med 7: e1000324.

Guillet, P., R. N'Guessan, F. Darriet, M. Traoreâ€ Lamizana, F. Chandre, and P. Carnevale. 2001. Combined pyrethroid and carbamate "two-in-one" treated mosquito nets: field efficacy against pyrethroid-resistant *Anopheles gambiae* and *Culex quinquefasciatus*. Med. Vet. Entomol. 15: 105–112.

Hougard, J. M., V. Corbel, R. N'Guessan, F. Darriet, F. Chandre, M. Akogbeto, T. Baldet, P. Guillet, P. Carnevale, and M. Traoré-Lamizana. 2003. Efficacy of mosquito nets treated with insecticide mixtures or mosaics against insecticide resistant *Anopheles gambiae* and *Culex quinquefasciatus* (Diptera: Culicidae) in Cote d'Ivoire. Bull. Entomol. Res. 93: 491–498.

Le Goff, G., F. M. Randimby, V. Rajaonarivelo, R. Laganier, J. M. Leong Pock Tsy, C. S. Ceianu, J. B. Duchemin, and R. Robert. 2003. *Anopheles mascarensis* de Meillon 1947, vecteur de paludisme dans le Moyen-Ouest de Madagascar? Arch. Inst. Pasteur Madagascar 69: 57–62.

Lepers, J. P., D. Fontenille, M. D. Rason, C. Chougnet, P. Astagneau, P. Coulanges, and P. Deloron. 1991. Transmission and epidemiology of newly transmitted falciparum malaria in the central highland plateaux of Madagascar. Ann. Trop. Med. Parasitol. 85: 297–304.

Nepomichene, T.N.J.J., E. Tata, and S. Boyer. 2015. Malaria case in Madagascar, probable implication of a new vector, *Anopheles coustani*. Malar. J. 14: 475.

Pates, H., and C. Curtis. 2005. Mosquito behavior and vector control. Annu. Rev. Entomol. 50: 53–70.

Pluess, B., F. C. Tanser, C. Lengeler, and B. L. Sharp. 2010. Indoor residual spraying for preventing malaria. Cochrane Database Syst. Rev. 14: CD006657.

Pock Tsy, J.M.L., J. B. Duchemin, L. Marrama, P. Rabarison, G. Le Goff, V. Rajaonarivelo, and V. Robert. 2003. Distribution of the species of the *Anopheles gambiae* complex and first evidence of *Anopheles merus* as a malaria vector in Madagascar. Malar. J. 2: 33.

Rakotomanana, F., I. Jeanne, J. Duchemin, V. Pietra, L. Raharimalala, M. Tombo, and F. Ariey. 2001. Approche géographique dans la lutte contre le paludisme dans la région des Hautes Terres Centrales à Madagascar. Arch. Inst. Pasteur Madagascar 67: 27–30.

Randriantsimaniry, D. 1995. Lutte antivectorielle dans l'épidémie de paludisme des plateaux de Madagascar. Santé 5: 392–396.

Ratovonjato, J., G. L. Goff, E. Rajaonarivelo, E. M. Rakotondraibe, and V. Robert. 2003. [Recent observations on the sensitivity to pyrethroids and DDT of *Anopheles arabiensis* and *Anopheles funestus* in the central Highlands of Madagascar; preliminary results on the absence of the kdr mutation in An. arabiensis]. Archives de l'Institut Pasteur de Madagascar 69: 63–69.

Ratovonjato, J., M. Randrianarivelojosia, M. E. Rakotondrainibe, V. Raharimanga, L. Andrianaivolambo, G. L. Goff, C. Rogier, F. Ariey, S. Boyer, and V. Robert. 2014. Entomological and parasitological impacts of indoor residual spraying with DDT, alphacypermethrin and deltamethrin in the western foothill area of Madagascar. Vectors 19: 20.

Romi, R., M. C. Razaiarimanga, R. Raharimanga, E. M. Rakotondraibe, L. H. Ranaivo, V. Pietra, A. Raveloson, and G. Majori. 2002. Impact of the malaria control campaign (1993-1998) in the highlands of Madagascar: Parasitological and entomological data. Am. J. Trop. Med. Hyg. 66: 2–6.

Scott, J. A., W. G. Brogdon, and F. H. Collins. 1993. Identification of single specimens of the Anopheles gambiae complex by the polymerase chain reaction. American Journal of Tropical Medicine and Hygiene 49: 520–529.

Tantely, M. L., J. C. Rakotoniaina, E. Tata, L. Andrianaivolambo, F. Razafindrasata, D. Fontenille, and N. Elissa. 2013. Biology of mosquitoes that are potential vectors of Rift Valley Fever virus in different biotopes of the Central Highlands of Madagascar. J. Med. Entomol. 50: 603–610.

Tantely, M. L., G. Le Goff, S. Boyer, and D. Fontenille. 2016. An updated checklist of mosquito species (Diptera: Culicidae) from Madagascar. Parasite 23: 20.

WHO 2006. Pesticides and their application: for the control of vectors and pests of public health importance. WHO, Geneva.

WHO 2007. WHO recommended insecticides for indoor residual spraying against malaria vectors. WHO Pesticides Evaluation Scheme (WHOPES), Geneva.

WHO 2012. World malaria report 2012, Geneva.

WHO 2013. Indoor Residual Spraying: An operational manual for indoor residual spraying (IRS) for malaria transmission control and elimination, Geneva.

WHO 2014. World malaria report 2013. World Health Organization, Geneva.

Downloaded from https://academic.oup.com/jme/article-abstract/54/4/1031/3068836
by University of Leicester user
on 19 January 2018